Two-Strain Invasion Model

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1 Two-Strain Competition and Invasion Model

To investigate the evolutionary dynamics of virulence (R_0), we extend the single-strain metapopulation framework to a two-strain competition model. We consider a resident strain (Strain 1) and an invading strain (Strain 2) with basic reproduction numbers R_{01} and R_{02}, respectively. To isolate the selection pressure acting purely on R_0, we assume both strains share the same per-flea infectivity, i.e., $\alpha_1 = \alpha_2 = \alpha$.

1.1 Mixed Flea Pool and Cross-Patch Transport

Let $S_{1,j}^{(k-1)}$ and $S_{2,j}^{(k-1)}$ be the number of hosts in patch j that died from Strain 1 and Strain 2 in the previous year, respectively. The total number of infectious carcasses (and thereby the source of the environmental flea pool) in patch j is:

$$S_j^{(k-1)} = S_{1,j}^{(k-1)} + S_{2,j}^{(k-1)}. \tag{1}$$

The proportions of the two strains in the local environmental flea pool are strictly determined by their respective death tolls:

$$q_{1,j} = \frac{S_{1,j}^{(k-1)}}{S_j^{(k-1)}}, \qquad q_{2,j} = \frac{S_{2,j}^{(k-1)}}{S_j^{(k-1)}}. \tag{2}$$

(If $S_j^{(k-1)} = 0$, we define $q_{1,j} = q_{2,j} = 0$).

Limited by the physical carrying capacity of migrating hosts, the expected total number of fleas exported from patch j ($E_j^{(k)}$) and retained locally ($R_j^{(k)}$) are:

$$E_j^{(k)} = \min\Big(\nu S_j^{(k-1)},\, \rho c_0 N_j^{(k,1)}\Big), \tag{3}$$

$$R_j^{(k)} = \nu S_j^{(k-1)} - E_j^{(k)}. \tag{4}$$

Consequently, the absolute number of fleas carrying strain $m \in \{1, 2\}$ that successfully arrive at the target patch i from both local retention and the global network is:

$$F_{m,i}^{(k)} = q_{m,i} R_i^{(k)} + \frac{1}{n} \sum_{j=1}^{n} q_{m,j} E_j^{(k)}. \tag{5}$$

Symbol	Definition	Units	Notes
$S_{i,j}^{(k)}$	host deaths from strain i in patch j in year k	number	
$S_j^{(k)} = \sum_i S_{i,j}^{(k)}$	total host deaths in patch j in year k	number	
$q_{i,j}$	proportions of strain i in patch j		
ν	scaling constant: epidemic size to infectious propagules		
ρ	infectious propagules per live host ($\approx$ flea capacity)		
r	host population growth rate		
K	host patch carrying capacity		same for all patches
c	host emigration probability		
$\mathcal{R}_0$			
D	infection fatality proportion		assumed = 1
α	transmission efficiency (successful infections) per propagule		

1.2 Poisson Colonization and Initial Seeds

Following the derivation of the per-host infection probability in the single-strain model, the expected number of infectious propagules arriving at patch i for each strain is proportional to the local flea pool and the per-flea infectivity α. However, to model the successful establishment of independent lineages (Galton-Watson processes), we must account for stochastic extinction (fizzle).

The probability that an initial infection escapes stochastic extinction in a supercritical branching process is $1 - 1/R_0$. Applying Poisson thinning, the effective expected number of *successfully established* initial seeds for each strain is:

$$\lambda_{1,i}^{(k)} = \alpha F_{1,i}^{(k)} \left(1 - \frac{1}{R_{01}}\right), \qquad \lambda_{2,i}^{(k)} = \alpha F_{2,i}^{(k)} \left(1 - \frac{1}{R_{02}}\right). \tag{6}$$

Since the initial per-host infection probability is small ($\pi \ll 1$), the colonization of the two strains can be modeled as independent Poisson sampling processes based on these thinned rates. This directly yields the integer number of successfully established initial seeds, $a_i^{(k)}$ and $b_i^{(k)}$, which will serve as the foundational lineages for the within-patch competition:

$$a_i^{(k)} \sim \text{Poisson}\left(\lambda_{1,i}^{(k)}\right), \qquad b_i^{(k)} \sim \text{Poisson}\left(\lambda_{2,i}^{(k)}\right). \tag{7}$$

1.3 Within-Patch Outbreak and Competition

To facilitate efficient numerical simulation across the metapopulation, we formulate the patch-level outbreak logic using indicator functions. We define the establishment indicator for each strain as:

$$I_{1,i}^{(k)} = \mathbb{1}_{\left\{a_i^{(k)}>0\right\}}, \qquad I_{2,i}^{(k)} = \mathbb{1}_{\left\{b_i^{(k)}>0\right\}}. \tag{8}$$

And the indicator for any outbreak occurring in the patch:

$$I_{\text{any},i}^{(k)} = \mathbb{1}_{\left\{a_i^{(k)}+b_i^{(k)}>0\right\}}. \tag{9}$$

When evaluating the macroscopic epidemic trajectory, the effective basic reproduction number in patch i depends on which strains successfully established. We assign $\bar{R}_0 = (R_{01} + R_{02})/2$ for co-infections:

$$R_{\text{eff},i}^{(k)} = R_{01} I_{1,i}^{(k)} \left(1 - I_{2,i}^{(k)}\right) + R_{02} \left(1 - I_{1,i}^{(k)}\right) I_{2,i}^{(k)} + \bar{R}_0 I_{1,i}^{(k)} I_{2,i}^{(k)}. \tag{10}$$

Let $z(R_0)$ denote the final epidemic size (as a proportion of the population), given by the non-zero root of $1 - z = \exp(-R_0 z)$. The total number of dead hosts in patch i can be computed concisely as:

$$Z_i^{(k)} = I_{\text{any},i}^{(k)} \cdot z\left(R_{\text{eff},i}^{(k)}\right) N_i^{(k,2)}. \tag{11}$$

During the early exponential growth phase, the strains behave as independent Galton-Watson branching processes. When the local host capacity is exhausted, the total available ecological niche ($Z_i^{(k)}$) is strictly partitioned proportionally to their initial seed sizes. This proportional allocation gracefully resolves both single-strain dynamics (where the fraction evaluates to 1) and co-infection competition:

$$S_{1,i}^{(k)} = Z_i^{(k)} \frac{a_i^{(k)}}{a_i^{(k)} + b_i^{(k)}}, \qquad S_{2,i}^{(k)} = Z_i^{(k)} \frac{b_i^{(k)}}{a_i^{(k)} + b_i^{(k)}}, \tag{12}$$

with the standard convention that if $a_i^{(k)} + b_i^{(k)} = 0$, both allocations evaluate to 0.

1.4 Host Ecological Update

Host population dynamics are blind to the specific strain causing mortality. The total death toll in patch i is updated as $S_i^{(k)} = S_{1,i}^{(k)} + S_{2,i}^{(k)}$. The surviving population $N_i^{(k,2)} - S_i^{(k)}$ then undergoes standard logistic growth to produce the initial host population $N_i^{(k+1,1)}$ for the following year.